

FULL NAME	USMAN IBRAHIM
REGISTRATION NUMBER	C11/26/FCDF/17159

LABORATORY REPORT ON OPNsense Firewall Policy Testing Logging and Packet Analysis

```
icdfa@icdfa-ns-lab-client-v1:~$ ip -4 -br address
lo                UNKNOWN          127.0.0.1/8
enp0s17           UP                10.10.10.163/24
icdfa@icdfa-ns-lab-client-v1:~$ ip route
default via 10.10.10.1 dev enp0s17 proto dhcp src 10.10.10.163 metric 100
10.10.10.0/24 dev enp0s17 proto kernel scope link src 10.10.10.163 metric 100
icdfa@icdfa-ns-lab-client-v1:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.
64 bytes from 1.1.1.1: icmp_seq=1 ttl=62 time=231 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=62 time=230 ms
64 bytes from 1.1.1.1: icmp_seq=3 ttl=62 time=231 ms
64 bytes from 1.1.1.1: icmp_seq=4 ttl=62 time=204 ms

--- 1.1.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 204.225/223.982/230.859/11.410 ms
icdfa@icdfa-ns-lab-client-v1:~$ getent hosts example.com
2606:4700:10::6814:179a example.com
2606:4700:10::ac42:93f3 example.com

icdfa@icdfa-ns-lab-client-v1:~$ curl -I http://example.com
HTTP/1.1 200 OK
Date: Tue, 06 Oct 2026 19:10:07 GMT
Content-Type: text/html; charset=utf-8
Connection: keep-alive
Server: cloudflare
Last-Modified: Sun, 04 Oct 2026 20:44:03 GMT
Allow: GET, HEAD
Accept-Ranges: bytes
Age: 10695
cf-cache-status: HIT
CF-RAY: a466f6e399a2668e-AMS
alt-svc: h3=":443"; ma=86400

icdfa@icdfa-ns-lab-client-v1:~$ curl -I https://example.com
HTTP/2 200
date: Tue, 06 Oct 2026 19:10:21 GMT
content-type: text/html; charset=utf-8
server: cloudflare
last-modified: Fri, 02 Oct 2026 16:11:02 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 10706
cf-cache-status: HIT
cf-ray: a466f7357c1b3581-AMS
alt-svc: h3=":443"; ma=86400

icdfa@icdfa-ns-lab-client-v1:~$
```

```
icdfa@icdfa-ns-lab-client-v1:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.

--- 1.1.1.1 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3100ms

icdfa@icdfa-ns-lab-client-v1:~$ curl -I https://example.com
HTTP/2 200
date: Tue, 06 Oct 2026 20:28:35 GMT
content-type: text/html; charset=utf-8
server: cloudflare
last-modified: Fri, 02 Oct 2026 16:11:02 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 1000
cf-cache-status: HIT
cf-ray: a46769d0394e2383-AMS
alt-svc: h3=":443"; ma=86400

icdfa@icdfa-ns-lab-client-v1:~$ getent hosts example.com
2606:4700:10::ac42:93f3 example.com
2606:4700:10::6814:179a example.com
icdfa@icdfa-ns-lab-client-v1:~$

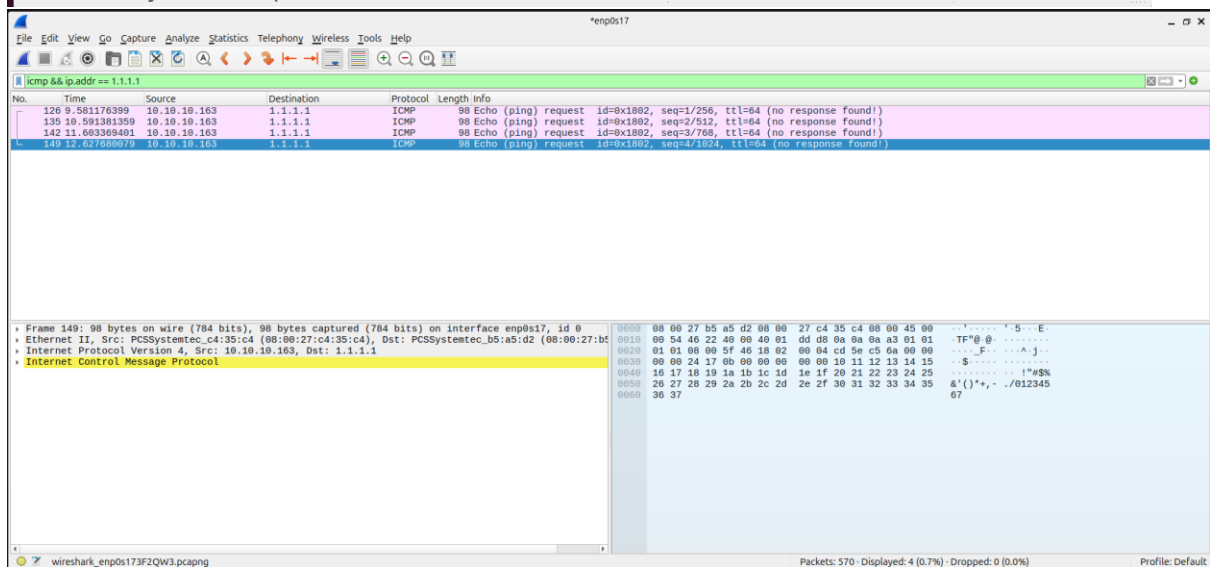
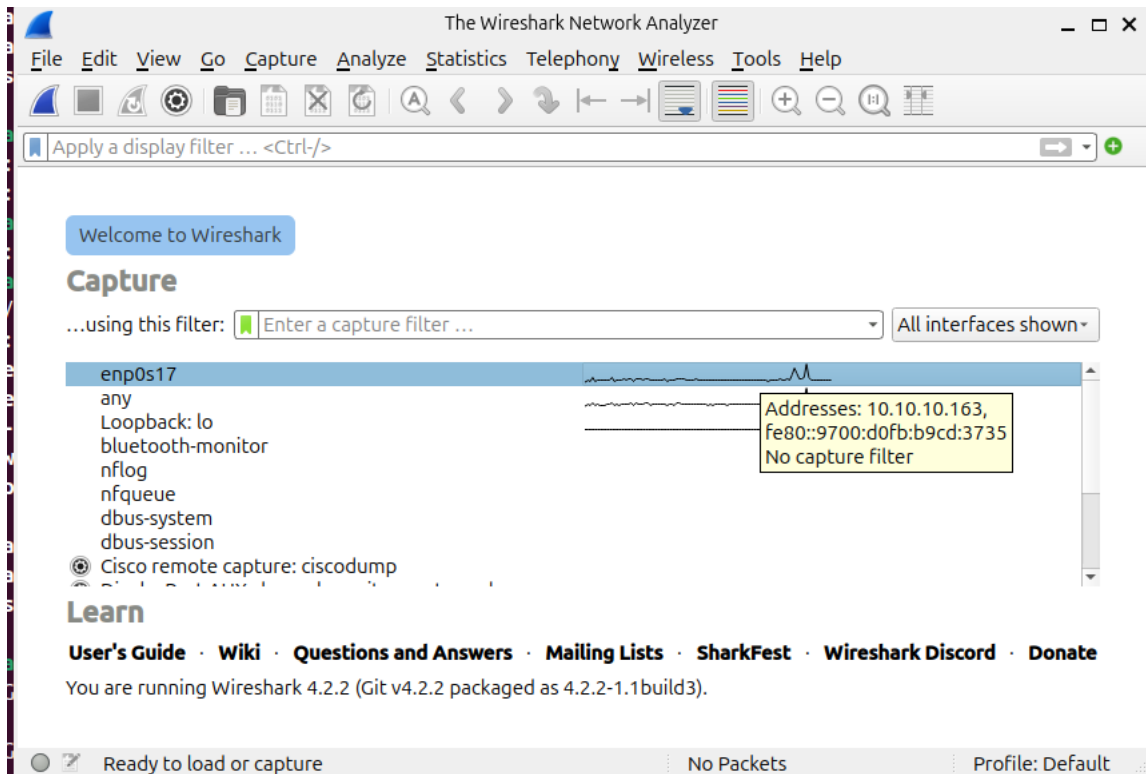
icdfa@icdfa-ns-lab-client-v1:~$ curl --max-time 10 -I http://example.com
curl: (28) Connection timed out after 10003 milliseconds
icdfa@icdfa-ns-lab-client-v1:~$ curl --max-time 10 -I https://example.com
HTTP/2 200
date: Tue, 06 Oct 2026 20:34:14 GMT
content-type: text/html; charset=utf-8
server: cloudflare
last-modified: Fri, 02 Oct 2026 16:11:02 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 1339
cf-cache-status: HIT
cf-ray: a4677216ec515d08-AMS
alt-svc: h3=":443"; ma=86400

icdfa@icdfa-ns-lab-client-v1:~$
```

The top screenshot displays the 'Firewall: Rules' configuration page in OPNsense. The left sidebar shows the navigation menu with 'Rules' selected. The main content area shows a table of firewall rules. The first rule is 'LAB2 BLOCK ICMP TO 1.1.1.1', which is enabled and has a priority of 1. The second rule is 'LAB2 BLOCK OUTBOUND HTTP', which is also enabled and has a priority of 1. The third rule is 'Default allow LAN to any rule', which is disabled and has a priority of 1. The fourth rule is 'Default allow LAN IPv6 to any rule', which is disabled and has a priority of 1. The table has columns for 'Version', 'Protocol', 'Source', 'Port', 'Destination', 'Port', 'Gateway', 'Description', and 'Commands'. Below the table, there is a blue bar with the text 'Apply. After changing settings, please remember to apply them.'

The bottom screenshot displays the 'Firewall: Log Files: Live View' page in OPNsense. The left sidebar shows the navigation menu with 'Log Files' selected. The main content area shows a table of firewall logs. The table has columns for 'Interface', 'Time', 'Protocol', 'Source', 'Destination', 'Action', and 'Label'. The logs show various traffic events, including blocked and passed traffic. The first log entry shows a blocked traffic event from 10.10.10.15 to 217.144.132.253:53. The second log entry shows a blocked traffic event from 10.10.10.163:5448 to 91.189.91.58:80. The third log entry shows a blocked traffic event from 10.10.10.163:5448 to 102.205.44.16.123. The fourth log entry shows a blocked traffic event from 10.10.10.163:5448 to 91.189.91.58:80. The fifth log entry shows a blocked traffic event from 10.10.10.163:5448 to 91.189.91.58:80. The sixth log entry shows a blocked traffic event from 10.10.10.163:5448 to 91.189.91.58:80. The seventh log entry shows a blocked traffic event from 10.10.10.163:5448 to 91.189.91.58:80. The eighth log entry shows a blocked traffic event from 10.10.10.163:5448 to 91.189.91.58:80. The ninth log entry shows a blocked traffic event from 10.10.10.163:5448 to 91.189.91.58:80. The tenth log entry shows a blocked traffic event from 10.10.10.163:5448 to 91.189.91.58:80. The eleventh log entry shows a blocked traffic event from 10.10.10.163:5448 to 91.189.91.58:80. The twelfth log entry shows a blocked traffic event from 10.10.10.163:5448 to 91.189.91.58:80. The thirteenth log entry shows a blocked traffic event from 10.10.10.163:5448 to 91.189.91.58:80. The fourteenth log entry shows a blocked traffic event from 10.10.10.163:5448 to 91.189.91.58:80. The fifteenth log entry shows a blocked traffic event from 10.10.10.163:5448 to 91.189.91.58:80. The sixteenth log entry shows a blocked traffic event from 10.10.10.163:5448 to 91.189.91.58:80. The seventeenth log entry shows a blocked traffic event from 10.10.10.163:5448 to 91.189.91.58:80. The eighteenth log entry shows a blocked traffic event from 10.10.10.163:5448 to 91.189.91.58:80. The nineteenth log entry shows a blocked traffic event from 10.10.10.163:5448 to 91.189.91.58:80. The twentieth log entry shows a blocked traffic event from 10.10.10.163:5448 to 91.189.91.58:80.

The traffic matched the block rule because OPNsense checks rules from top to bottom. The block rule was placed above the default allow rule, so the firewall blocked the traffic before it could be allowed.



Wireshark packet capture analysis showing network traffic details, packet list, packet details, and packet bytes.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
270	31.841694913	10.10.10.103	104.20.23.154	TCP	74	[TCP Retransmission] 53122 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=364413818 TSecr=0 WS=1024
270	32.852301408	10.10.10.103	104.20.23.154	TCP	74	[TCP Retransmission] 53122 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=364413818 TSecr=0 WS=1024
280	33.875317233	10.10.10.103	104.20.23.154	TCP	74	[TCP Retransmission] 53122 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=364413818 TSecr=0 WS=1024
290	34.88122635	10.10.10.103	104.20.23.154	TCP	74	[TCP Retransmission] 53122 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=364413818 TSecr=0 WS=1024
300	35.924119601	10.10.10.103	104.20.23.154	TCP	74	[TCP Retransmission] 53122 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=364413818 TSecr=0 WS=1024
380	36.846954107	10.10.10.103	172.66.147.243	TCP	74	[TCP Retransmission] 39898 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=899293496 TSecr=0 WS=1024
310	37.889861201	10.10.10.103	172.66.147.243	TCP	74	[TCP Retransmission] 39898 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=899293496 TSecr=0 WS=1024
327	38.931104241	10.10.10.103	172.66.147.243	TCP	74	[TCP Retransmission] 39898 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=899293496 TSecr=0 WS=1024
334	39.955871534	10.10.10.103	172.66.147.243	TCP	74	[TCP Retransmission] 39898 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=899293496 TSecr=0 WS=1024
343	40.979576251	10.10.10.103	172.66.147.243	TCP	74	[TCP Retransmission] 39898 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=899293496 TSecr=0 WS=1024

Packet Details:

Frame 270: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface enp0s17, id 0

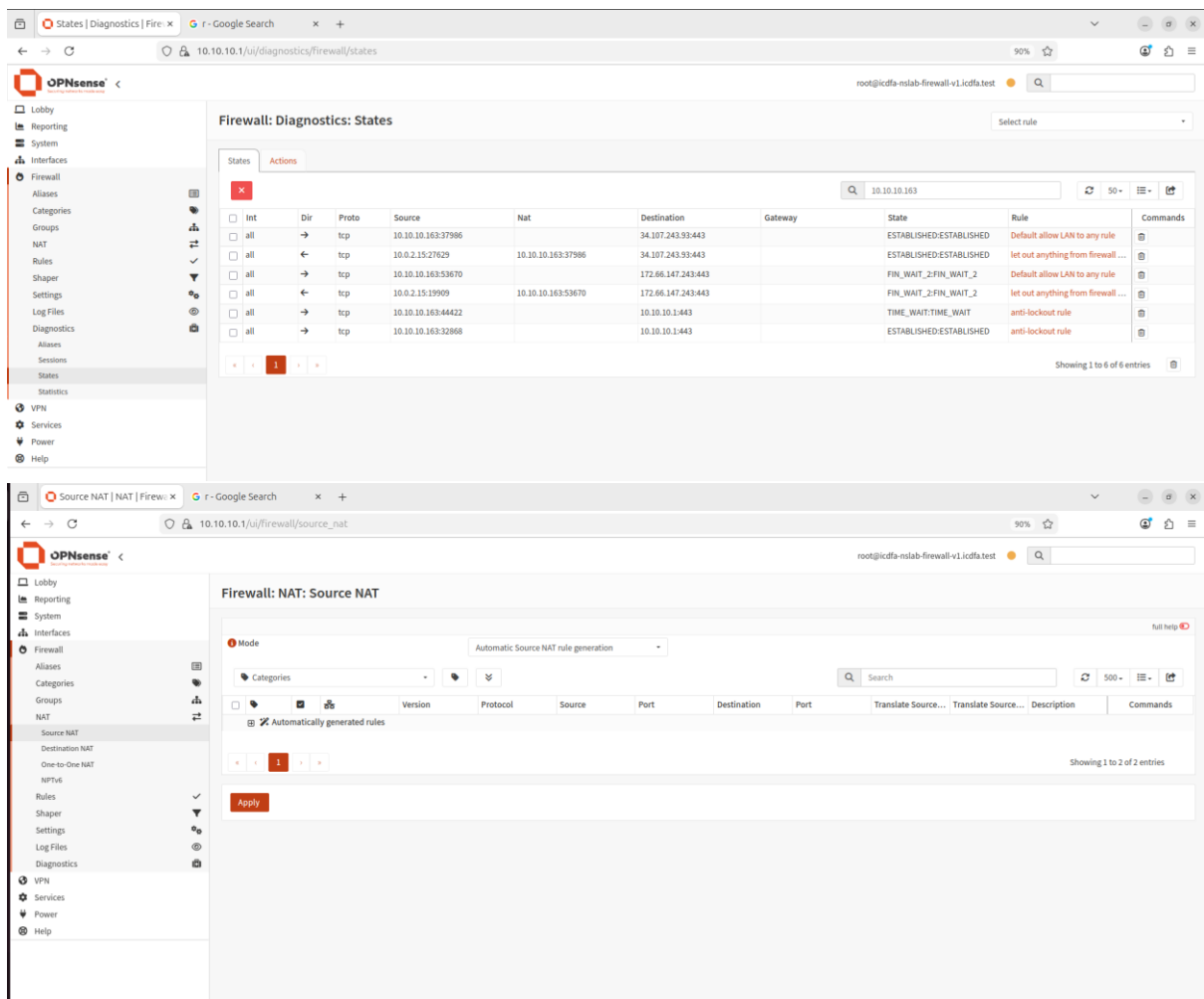
Ethernet II, Src: PCSSystemtec_c4:35:c4 (08:00:27:c4:35:c4), Dst: PCSSystemtec_b5:a5:d2 (08:00:27:b5:a5:d2)

Internet Protocol Version 4, Src: 10.10.10.103, Dst: 104.20.23.154

Transmission Control Protocol, Src Port: 53122, Dst Port: 80, Seq: 0, Len: 0

Packet Bytes:

0000 08 00 27 b5 a5 d2 08 00 27 c4 35 c4 08 00 45 005...E
0010 00 3c 3b 71 40 00 40 06 6a f0 0a 0a 0a a3 68 14 <:q0 @ j.....h
0020 17 9a cf 82 00 50 34 2c 52 ff 00 00 00 00 a0 82P4, R.....
0030 fa f0 04 89 00 00 02 04 05 b4 04 02 00 0a 15 b8
0040 83 7a 00 00 00 00 01 03 03 0a2.....



The NAT mode was set to Automatic Outbound NAT. This allows OPNsense to translate private IP addresses from the 10.10.10.0/24 network to the WAN address for Internet access.

```
icdfag@icdfa-ns-lab-client-v1:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data:
64 bytes from 1.1.1.1: icmp_seq=1 ttl=62 time=195 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=62 time=214 ms
64 bytes from 1.1.1.1: icmp_seq=3 ttl=62 time=192 ms
64 bytes from 1.1.1.1: icmp_seq=4 ttl=62 time=215 ms

--- 1.1.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 192.248/204.211/214.836/10.465 ms
icdfag@icdfa-ns-lab-client-v1:~$ curl --max-time 10 -I http://example.com
HTTP/1.1 200 OK
Date: Tue, 06 Oct 2026 21:12:00 GMT
Content-Type: text/html; charset=utf-8
Connection: keep-alive
Server: cloudflare
Last-Modified: Fri, 02 Oct 2026 16:11:02 GMT
Allow: GET, HEAD
Accept-Ranges: bytes
Age: 3615
cf-cache-status: HIT
CF-RAY: a467a99d1994fb81-AWS
alt-svc: h3=":443"; ma=86400

icdfag@icdfa-ns-lab-client-v1:~$ curl --max-time 10 -I https://example.com
HTTP/2 200
date: Tue, 06 Oct 2026 21:12:20 GMT
content-type: text/html; charset=utf-8
server: cloudflare
last-modified: Fri, 02 Oct 2026 16:11:02 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 3626
cf-cache-status: HIT
cf-ray: a467a9eb107726ec-AWS
alt-svc: h3=":443"; ma=86400
```

Questions and Answers

Why must the specific block rules be placed above the broad allow rule?

OPNsense checks firewall rules from top to bottom. If the allow rule is above the block rule, the traffic will be allowed before it reaches the block rule.

Which five packet attributes are most useful when explaining a firewall decision?

1. Source IP address
2. Destination IP address
3. Protocol (TCP, UDP, ICMP)
4. Source port
5. Destination port

Why did blocking ICMP not block HTTPS?

Because ICMP and HTTPS are different protocols. Blocking ICMP only stops ping traffic, while HTTPS traffic is still allowed.

What difference did you observe between the blocked TCP port 80 traffic and permitted TCP port 443 traffic?

Traffic on port 80 failed and timed out because it was blocked. Traffic on port 443 worked normally because it was allowed.

What role does outbound NAT play when icdfa-nslab-client-v1 uses a private IPv4 address?

Outbound NAT changes the private IP address to the firewall's WAN address so the computer can access the Internet.

Why is restoring the original state an important part of a controlled security laboratory?

Restoring the original state makes sure the network works normally again and prepares the system for the next lab or activity.

SUMMARY

In this lab, I tested firewall rules using OPNsense and Ubuntu. First, I checked that the network, internet, DNS, HTTP, and HTTPS were working properly.

I then created a rule to block ping (ICMP) to 1.1.1.1 and confirmed that ping stopped working while DNS and HTTPS still worked.

Next, I created another rule to block HTTP (port 80) traffic. HTTP stopped working, but HTTPS (port 443) continued to work.

I checked the firewall logs and Wireshark captures to see the blocked and allowed traffic. I also verified that NAT was translating the private IP address for internet access.

Finally, I disabled the test rules and confirmed that ping, HTTP, and HTTPS were working again. This showed that the firewall rules worked correctly and that normal connectivity was successfully restored.